MD HASIBUL HOSSAIN

Civil Engineer · Researcher in Sustainable Construction Materials and Environmental Assessment

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Khilkhet, Dhaka-1229, Bangladesh

Nationality: Bangladeshi
Date of birth: 3 March 1998
Place of birth: Kishoreganj

PERSONAL STATEMENT

Civil Engineering graduate with four years of laboratory research training across structural, geotechnical and environmental engineering, complemented by five years of professional practice in sustainability assessment, construction supervision and management-system auditing. My undergraduate thesis on rice husk ash as a partial cement replacement established, over 54 concrete specimens, that 5 % replacement raises 28-day compressive strength by 21.8 % while lowering material cost; earlier work applied artificial neural network models to forecast industrial impacts on urban river water quality. Together these established the research direction I intend to pursue at postgraduate level.

Professionally I have progressed from site engineering to supervising multi-site infrastructure projects, while conducting ISO 9001, 14001 and 45001 audits, energy and ETP assessments, GRI-standard sustainability reporting, and cadastral GIS digitisation for national land-zoning work. I am now seeking admission to a Master's programme in Environmental Science or a closely related field, with a research focus on sustainable construction materials, environmental impact assessment and climate-resilient urban development.

EDUCATION

B.Sc. in Civil Engineering
2018 – June 2022

Presidency University, Dhaka, Bangladesh

Major: Structural Engineering · Minor: Environmental Engineering · Final grade: 3.42 / 4.00

- Thesis: Utilisation of Rice Husk Ash (RHA) as Partial Replacement of Cement for Sustainable Concrete — group thesis of six authors, supervised by Susmita Majumder.

Higher Secondary Certificate (HSC), Science
2014 – 2016

Cambrian School & College, Dhaka

Final grade: 4.75 / 5.00

Secondary School Certificate (SSC), Science
2012 – 2014

Asiya Bari Ideal School, Kishoreganj

Final grade: 4.88 / 5.00

RESEARCH EXPERIENCE

Undergraduate Researcher, Structural Engineering Laboratory
Apr 2021 – Jun 2022

Presidency University · Supervisor: Susmita Majumder

- Cast, cured and compression-tested 54 cylindrical specimens across six cement-replacement levels (0–25 %) and three curing ages (7, 14 and 28 days) at a 1 : 1.5 : 3 mix ratio, W/C 0.45.
- Established that 5 % replacement raises 28-day compressive strength by 21.8 % (2,724 to 3,317 psi), with strength declining sharply beyond 10 % replacement.
- Characterised the ash as strongly siliceous (62.3 % SiO_2 against 20.5 % for the cement replaced), confirming the pozzolanic basis of the gain.
- Quantified the trade-off: material cost fell by up to 6.3 % while water absorption rose from 2.63 % to 4.32 %, fixing the practical window at 5–10 % replacement.

Undergraduate Researcher, Geotechnical Engineering Laboratory
Mar 2020 – Jan 2021

Presidency University · Supervisor: Prof. Dr. Tushar Ahamed

- Designed custom instrumentation where standard apparatus did not cover the required measurements, and conducted ASTM-standard geotechnical testing.
- Developed column behaviour models in PLAXIS 3D, simulating load transfer, deformation and failure mechanisms of granular pile systems.
- Analysed experimental and numerical data to assess technical and economic feasibility.
- Trained and mentored undergraduate students in experimental design, data analysis and numerical modelling.

Undergraduate Researcher, Geotechnical Engineering Laboratory
Feb 2019 – Jan 2020

Presidency University · Supervisor: Prof. Dr. Tushar Ahamed

- Determined soil parameters through ASTM-compliant laboratory testing using purpose-designed instrumentation; measured values, not literature defaults, formed the model input.
- Built PLAXIS 3D finite element models simulating reclaimed asphalt pavement (RAP)-infused granular piles under load.
- Conducted literature reviews and contributed to the thesis proposal framing the second phase of the programme.

Undergraduate Researcher, Environmental Engineering Laboratory
Jun 2018 – Dec 2018

Presidency University · Supervisor: Assoc. Prof. Dr. Asma Akter

- Developed artificial neural network models to forecast industrial impacts on urban river water quality.
- Compiled 20 years of surface-water quality records through coordination with government agencies, reconciling differing formats and sampling regimes.
- Performed time-series and correlation analyses to identify the governing pollution parameters.

PROFESSIONAL EXPERIENCE

Project Supervisor
Apr 2025 – Present

Plan Bangla Consultant Firm (PBCF), Bangladesh

- Supervise concurrent civil and infrastructure projects, conducting technical inspections at each construction stage against design specifications, BNBC provisions and client quality requirements.
- Lead multidisciplinary site teams of engineers, subcontractors and skilled labour, allocating manpower, sequencing works and resolving on-site technical issues.
- Prepare inspection reports, progress documentation and technical recommendations supporting decisions on remedial and retrofitting works.
- Enforce occupational health and safety protocols aligned with ISO 45001 principles, and environmental safeguards including construction waste management, dust and silt-runoff control, and Department of Environment compliance.
- Carry out cadastral GIS digitisation for national land-zoning work: georeferencing and spatial adjustment of Mouza sheets, edge-matching, vectorisation and topological correction in ArcGIS.

Sustainability Engineer
Sep 2021 – Jan 2025

The Sustainability Nexus Ltd. (SusNex), Uttara, Dhaka

- Conducted management system audits (ISO 9001, ISO 14001, ISO 45001), energy assessments, ETP assessments and Detailed Engineering Assessments for industrial clients.
- Prepared GRI-standard sustainability and ESG reports, and carried out material traceability and circular economy research.
- Contributed to business development, proposal writing and training management.

Structural Engineering Project Supervisor (Independent)
2022 – Jul 2024

Independent practice, Bangladesh

- Oversaw the design and construction of seven two-storey residential buildings.
- Managed project planning, structural design and site supervision, delivering on schedule and within budget.

Instructor: Physics, Chemistry and Biology
2022 – Mar 2025

Upayan Coaching Center, Khilkhet, Dhaka

- Taught science at secondary and higher-secondary level, emphasising conceptual clarity and examination strategy.
- Designed lesson plans developing critical thinking and mentored students preparing for competitive admission examinations.

Consultant, Structural Engineering
May 2021 – Nov 2021

Home Haven Design and Consultancy, Dhaka

- Advised on project designs for compliance with structural integrity standards and local building regulations.
- Minimised environmental impact through construction waste management, control of air and water pollution, and DoE compliance.

Site Engineer (Voluntary)
2019 – Mar 2021

Various construction companies, Bangladesh

- Managed on-site construction activities, overseeing safety protocols and quality control.

SELECTED ASSESSMENT AND REPORTING PROJECTS

Year	Project	Client / partner	Role
2024	Detailed Engineering Assessment (DEA) and retrofitting design	Teen Age Modern Fashion Ltd.	Engineer, Civil
2024	Assessment of NOSTRI Fire Safety Lab requirements and architectural design	GIZ (German Development Agency)	Sustainability Engineer, Civil
2023–24	Environment sustainability and climate change assessment	The Fred Hollows Foundation	Sustainability Engineer
2022–24	ESG data reporting and GRI-standard sustainability reports for four member factories	BGMEA	Sustainability Developer
2023	Internal management system audit, ISO 9001:2015	The Sustainability Nexus Ltd.	Internal Auditor
2023	Research support on the JHUT (textile waste) market system of Bangladesh	GIZ (German Development Agency)	Sustainability Developer
2023	Energy audit and assessment of electrical and mechanical equipment	Pioneer Group	Sustainability Engineer
2022–23	Sustainability strategy and sustainability report	Bashundhara Group	Sustainability Developer

DESIGN, DRAFTING AND MAPPING PORTFOLIO

Full portfolio with drawings and renders: mdhasibulhossain.com/portfolio

Architectural design	Five-storey residential building; two-storey duplex; single-storey residence; contemporary villa with gatehouse; Eidgah prayer ground and gateway. 2D planning through 3D modelling to presentation renders.	AutoCAD, SketchUp
Infrastructure and planning	Institutional campus master plan with playing field, canal crossings and road network; multi-span girder bridge with deck, pier layout, footway and lighting.	SketchUp
Working drawings	Dimensioned floor plans, sanitary and kitchen fixture layouts, furniture plans, and electrical sets covering conduit runs, light-load distribution and fitting positions.	AutoCAD
Cadastral GIS mapping	Georeferencing and spatial adjustment of Mouza sheets against satellite imagery, edge-matching adjacent sheets, vectorising plots, roads and rivers, and topological correction into clean geodatabases for national land-zoning.	ArcGIS, ArcMap, Google Earth

TECHNICAL SKILLS

Geotechnical testing (ASTM)	Particle Size Distribution (D7928) · Hydrometer Analysis (D7928) · Liquid Limit & Plasticity Index (D4318) · Specific Gravity of Soil Solids (D854-23) · Standard Proctor Compaction (D698) · Unconfined Compressive Strength (D2166) · Direct Shear (D3080) · Triaxial Compression (D4767) · California Bearing Ratio (CBR)
Materials and aggregate (ASTM)	Sieve Analysis, Fine & Coarse Aggregates (C136) · Absorption, Coarse Aggregates (C127) · Bulk Specific Gravity (C127) · Unit Weight and Voids in Aggregate (C29)
Engineering software	Bentley PLAXIS 3D · CSI ETABS, SAP, SAFE · Bentley STAAD Pro · AutoCAD · SketchUp · Fusion 360 · Lumion · Origin Pro
GIS	ArcGIS · ArcMap · Google Earth · georeferencing · spatial adjustment · edge-matching · vectorisation · topology correction · geodatabase construction
Modelling and analysis	Artificial neural network modelling · time-series analysis · correlation analysis · finite element modelling · C · C++ · Microsoft Excel (advanced)
Management systems	ISO 9001:2015 internal auditing · ISO 14001 and ISO 45001 audit practice · GRI Standards sustainability and ESG reporting · energy audits · ETP assessment · Detailed Engineering Assessment · retrofitting design review
Design and media	Adobe Photoshop, Illustrator, Lightroom, After Effects, XD, Audition · Filmora · Clipchamp · web development

RESEARCH INTERESTS

Sustainable construction materials · environmental impact assessment · ground improvement · granular pile foundations · finite element modelling · reclaimed asphalt pavement · geotextiles · artificial neural networks · foundation engineering · climate-resilient urban development

LANGUAGES

English — IELTS Academic, 23 November 2024 Overall 7.5

Listening 7.5 · Reading 7.5 · Speaking 8.0 · Writing 7.5

Bangla — mother tongue

Hindi — working proficiency in spoken interaction and listening comprehension

VOLUNTEERING AND HONOURS

United Nations Volunteers	Volunteer · January 2019 – February 2022
Bidyanondo Foundation	Volunteer · January 2020 – December 2024
Community Health Camp	Organiser · March 2021, Kishoreganj. Organised and managed a health camp providing medical services and health education to a rural community.
Honours	Award for contribution to social welfare projects, Social Services Club, Presidency University · Captain, Civil 2020 Cricket Team — runners-up, 2020 inter-university tournament · Member, PU Civil Badminton Team — champions, 2021 · Designer, Civil Fest 2020 (CESA)

REFERENCES

Dr. Susmita Majumder — info.ieltsroad@gmail.com
Founder & Chairman, IELTS Road; former Assistant Professor and thesis supervisor, Department of Civil Engineering, Presidency University

Dr. Tushar Ahamed — tusharahamed1973@gmail.com
Professor, School of Engineering, Primeasia University; former Associate Professor, Faculty of Engineering, Presidency University

Dr. Asma Akter — asmaakter4459@gmail.com
Consultant and Flood Modelling Expert, UN World Food Programme; former Assistant Professor, Faculty of Engineering, Presidency University

Mohammad Hafijur Rahaman — mr114@uark.edu
Research Fellow, University of Arkansas, USA; former Assistant Professor, Faculty of Engineering, Presidency University